

Roll No. ....

**TBT-503**

**B. TECH. (BIOTECHNOLOGY)  
(FIFTH SEMESTER)**

**MID SEMESTER EXAMINATION, Oct., 2022**

**BIOINFORMATICS**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define Bioinformatics in your own words. Also write any *five* applications of bioinformatics in Biotechnology. 10 Marks (CO5)

OR

- (b) Elaborate various operating systems (at least *two*) that are prevalent in use by bioinformatics softwares. 10 Marks (CO1)

2. (a) Differentiate between system software and application software with example. 10 Marks (CO1)

OR

- (b) Give a historical account and also contribution of various scientists in bioinformatics. 10 Marks (CO1)

**P. T. O.**

(2)

3. (a) What are databases ? What is the need of databases in Biotechnology ?  
Substantiate your answer with example of use of databases in at least  
*two* Biotechnology problems. 10 Marks (CO2)

OR

- (b) Differentiate primary and secondary databases along with examples.  
10 Marks (CO2)

4. (a) Elaborate UNIPROT and also various annotations provided in it for any  
entry. 10 Marks (CO2)

OR

- (b) What is NCBI ? How can it be of use in Biotechnology research ?  
10 Marks (CO1 and CO5)

5. (a) What do you understand by sequence alignment ? What do you  
understand by identity conserved substitution and semi-conserved  
substitution during sequence alignment ? 10 Marks (CO2 and CO5)

OR

- (b) What is INSDC ? What are its various members ? Also elaborate why it  
was formed. 10 Marks (CO2)